Computing Project
PROJ-H402

Lie Detector for LLMs

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Abstract

Large Language Models (LLMs) frequently produce factually incorrect statements, stemming from two distinct failure modes : hallucination (an honest mistake driven by missing knowledge) and deception (outputting a falsehood despite internally representing the truth). Distinguishing between the two requires bypassing the model's textual output to inspect its internal representations. Recent advances in Representation Engineering have shown that in massive models like Llama-2-13B, a linearly separable "truth direction" can be extracted from hidden states and generalized across diverse datasets. However, it remains an open question whether this property is an emergent consequence of massive scale or a fundamental structural feature of the Transformer architecture. This project investigates this question by reproducing a foundational truth-probing protocol on a much smaller, non-conversational base model : Microsoft Phi-2 (2.7B parameters). By standardizing 14 diverse datasets into a unified contrastive pipeline, extracting last-token hidden states across all 32 layers, and training four linear probe families (DIM, LR, PCA-G, LAT), we aim to determine if a universal, out-of-distribution truth direction can still be reliably recovered at a 5x smaller scale.

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1 Introduction

1.1 Motivation : When AI Outputs Are Wrong

Modern Large Language Models such as GPT-4, Claude or Gemini have reached a level of fluency at which a casual user can no longer reliably distinguish a factually correct statement from a confidently-asserted falsehood. The surface output of the model is, by construction, indistinguishable in form between a sincere statement, a mistake, and a deliberate misrepresentation. As LLMs are deployed in increasingly high-stakes contexts, legal drafting, medical triage, financial analysis the inability to audit the truthfulness of their outputs becomes an actionable safety problem.

1.2 Hallucination versus Lying

The literature on LLM failure modes distinguishes between two qualitatively different phenomena that share the same visible symptom :

— **Hallucination** : an "honest mistake" caused by missing knowledge, poor retrieval, or a faulty inference chain. The model outputs a falsehood because it does not internally represent the correct fact.

— **Lying** (or *deception*) : the model internally encodes the correct fact, but its output deviates from that internal representation. The decision pipeline can be schematised in four steps : (i) the model possesses the true fact, (ii) it internally verifies the fact, (iii) it elects to deceive, (iv) it emits the false statement.

From the outside, the so-called *black-box* view, these two cases are indistinguishable. Both produce an incorrect textual output and both can be expressed with high confidence. Detecting deception therefore requires a *white-box* methodology : bypassing the textual output and reading the model's internal state directly.

1.3 The White-Box Approach

The core insight of *Representation Engineering* is that high-level semantic concepts, including truthfulness, are often *linearly* encoded in the hidden states of large Transformers. If a concept admits a linear representation, then a low-capacity probe such as a logistic regression or a difference-in-means classifier suffices to recover it. Recovering a "truth direction" from hidden states would, in turn, provide an internal signal of whether the model itself classifies a candidate statement as true or false, regardless of what it ultimately emits.

1.4 Research Question

This project addresses a single, precise research question :

> *Does Microsoft Phi-2 (2.7B parameters) contain a reusable linear "truth direction" that generalises across datasets, or are truth probes mostly dataset-specific on a smaller base model ?*

The reference study by Wagner answers this question affirmatively for Llama-2-13B. By reproducing the same protocol on a model that is approximately 5$\times$ smaller, we test whether the result is robust to a substantial reduction in scale, or whether it is in fact an emergent property of large models that disappears below a certain parameter threshold.

1.5 Contributions

The contributions of this work are the following :

1. A faithful reproduction of the Wagner protocol on Microsoft Phi-2, with a unified pipeline that ingests 14 standardised datasets (12 Hugging Face, 2 local) into a single contrastive schema.
2. A 32-layer sweep characterising the emergence of the truth direction along the depth of Phi-2.
3. A full 7×7 out-of-distribution transfer matrix for four probe families (DIM, LR, PCA-G, LAT).

4. A side-by-side numerical comparison with Llama-2-13B that supports the hypothesis that the truth direction is a structural property of the Transformer architecture rather than an emergent property of scale.
5. An open-source implementation, released at `https://github.com/imadsharof/Lie-Detector-for-LLM`.

1.6 Report Structure

Chapter 2 reviews the relevant literature on Representation Engineering, linear probing and the original truth-probe study. Chapter 3 formalises the methodology : dataset standardisation, prompt template, activation extraction and the four probe families. Chapter 4 describes the implementation. Chapter 5 reports the experimental results. Chapter 6 discusses the results in comparison with the Llama-2-13B reference. Chapter 8 states the limitations and proposes future work. Chapter 9 concludes.

2 | Background and Related Work

2.1 Representation Engineering

Representation Engineering (RepEng) is a recent line of work that approaches the problem of interpretability and control of LLMs by treating their hidden representations as the primary object of study, rather than the textual output. The central empirical observation is that a wide range of high-level semantic concepts, truthfulness, sentiment, harmfulness, refusal, are encoded in the residual stream of large Transformers in a form that is approximately linear. As a consequence, a concept can be *read* from the hidden states by a low-capacity probe, and (more strikingly) can be *written into* the residual stream as a steering vector that modifies subsequent generation.

2.2 Linear Probing of Truth

The idea of probing a neural representation by training a small classifier on its activations predates the Transformer era. Within the LLM context, two recent results frame the present project :

— Burns et al. introduce Contrast Consistent Search (CCS), an *unsupervised* probe that searches for a direction in activation space along which contradictory statements consistently disagree.

— Marks and Tegmark show that a simple **difference-in-means** direction, computed on labelled true/false statements, recovers a truth representation that is competitive with, and often better than, more elaborate probes. They argue that truth is encoded along a single dominant axis in the residual stream of large LLMs.

2.3 The Generalisation Study of Wagner (2024)

The blog post and code base by Wagner (handle `mishajw`) ties these threads together. Using Llama-2-13B as a base model, Wagner trains linear probes on one dataset, evaluates them on every other dataset, and reports the resulting out-of-distribution transfer accuracies. The central finding is that the truth direction generalises far better than one would naively expect : a probe trained on DBpedia topic classification can decide truthfulness on Amazon sentiment with accuracy close to in-distribution performance. The present project is a direct, model-substituted reproduction of that study.

2.4 Why Phi-2

Microsoft Phi-2 is a 2.7B-parameter *base* (non-instruction-tuned) Transformer model with 32 layers and a hidden dimension of 2560. It is approximately 5$\times$ smaller than Llama-2-13B (40 layers, $\approx$ 13B parameters) and was selected for three reasons :

1. **Base model** : unlike Phi-3 or instruction-tuned variants, Phi-2 is trained without RLHF on top, so its representations are not distorted by post-training alignment. This matches the Llama-2 *base* model used in the reference study.
2. **Public weights, modest VRAM** : Phi-2 fits comfortably on a single consumer GPU, making the reproduction feasible without large-scale infrastructure.
3. **Smallest credible scale** : at 2.7B parameters Phi-2 is small enough to meaningfully stress-test the scale-versus-architecture hypothesis, while remaining capable enough to plausibly encode factual structure.

3 | Methodology

3.1 Pipeline Overview

The end-to-end experimental pipeline consists of five stages, executed sequentially :

1. **Dataset ingestion** : 14 datasets are loaded and converted into a single contrastive schema.
2. **Prompt construction** : each (question, candidate answer) pair is wrapped in the same declarative template.
3. **Activation extraction** : Phi-2 is run in inference mode with `output_hidden_states=True`; the hidden state at the last token is extracted for every layer.
4. **Probe training** : one of four probe families is fitted on the training split of a source dataset.
5. **Out-of-distribution evaluation** : the probe is evaluated on the test split of every other dataset, producing a transfer matrix.

3.2 Dataset Preparation

3.2.1 Sources

The 14 datasets cover four distinct linguistic regimes : sentiment classification, topic classification, multiple-choice question answering, and self-report truthfulness. Table 3.1 lists every source and its Hugging Face identifier (or local origin).

#	Dataset	Source	Type
1	`imdb`	`stanfordnlp/imdb`	Binary sentiment
2	`amazon_polarity`	`fancyzhx/amazon_polarity`	Binary sentiment
3	`ag_news`	`fancyzhx/ag_news`	4-way topic classification
4	`dbpedia_14`	`fancyzhx/dbpedia_14`	14-way topic classification
5	`rte`	`nyu-mll/glue` (`rte`)	Textual entailment
6	`boolq`	`google/boolq`	Boolean QA
7	`arc_easy`	`allenai/ai2_arc` (Easy)	Science QA
8	`arc_challenge`	`allenai/ai2_arc` (Chall.)	Science QA
9	`openbookqa`	`allenai/openbookqa`	Science QA
10	`commonsense_qa`	`tau/commonsense_qa`	Commonsense QA
11	`piqa`	`ybisk/piqa`	Physical reasoning
12	`truthful_qa`	`truthfulqa/truthful_qa`	Adversarial misconceptions
13	`facts`	Local, procedural	City/country + numeric contrasts
14	`repeng_truthful`	Local JSONL (`repeng`)	Honest/dishonest self-reports

Table 3.1 – The 14 datasets ingested by the pipeline. 12 are pulled from the Hugging Face Hub via `datasets.load_dataset`, 2 are constructed locally.

3.2.2 Contrastive Grouping

A core engineering decision is that every example is represented as a *group* of candidate answers sharing a single context. Each group is identified by a `group_id` and contains at least one candidate labelled true and at least one labelled false. This contrastive structure has two consequences :

— It forces probes to learn the *difference* between true and false activations within the same context, rather than exploit surface features of the input text.

— It enables a leakage-free split : when the dataset is partitioned into train/validation/test, the partition is performed at the **group level**. All candidate answers of a given question are kept in the same split.

The default budget is 80 groups per dataset. The split ratios are 60% train, 20% validation, 20% test.

3.2.3 Unified Schema

After ingestion, every example is represented as a single row of a Pandas DataFrame with the following six columns :

`dataset_name | group_id | question | answer | prompt | label`

3.3 The Prompt Template

A single declarative template is applied to every (question, answer) pair, regardless of the source dataset. The template is defined once in the codebase (`datasets.py`, line 27) and is the sole interface between the raw data and the model :

Listing 3.1 – The Phi-2 prompt template applied uniformly across all 14 datasets.

```
Task: Decide whether the candidate answer is correct.

Question:
{question}

Candidate answer:
{answer}

Correctness:
```

Phi-2 is a *base* model, not a chat model. The template therefore contains no system/user/assistant role markers, and ends with the open-ended cue `"Correctness:"` which forces the model to compress its factual assessment into the hidden state of the final token.

3.4 Activation Extraction

For each prompt, Phi-2 is run once in inference mode with `output_hidden_states=True`. The residual-stream activation at the **last token** (the colon of `"Correctness:"`) is extracted at every layer $\ell \in \{0, 1, \ldots, 31\}$. The last-token strategy is motivated by two complementary observations :

— In autoregressive Transformers, the last-token hidden state is the only state that has attended to the entire prompt and is about to drive next-token logits. It therefore acts as an information bottleneck.

— The literature on RepEng consistently identifies the last-token residual as the locus where high-level semantic concepts are most cleanly encoded.

Extracted activations are cached on disk as compressed NumPy arrays (`.npz`), keyed by the SHA-1 of the prompt text. The cache amortises the dominant computational cost across all downstream experiments : once activations are extracted, probe training is essentially free.

3.5 The Four Probe Families

Four linear probe families are evaluated, covering both supervised and weakly supervised regimes. Each probe ultimately produces a *direction* $\mathbf{w} \in \mathbb{R}^d$ in the hidden state space, and a candidate is classified by the sign of $\mathbf{w}^\top \mathbf{h}$.

Difference in Means (DIM, supervised) : The Difference in Means probe is the most intuitive and geometrically straightforward approach to finding a concept vector. Following the methodology of Marks and Tegmark, it computes the geometric center of mass for all activations corresponding to true statements, and subtracts the center of mass of all false statements.

$$\mathbf{w}_{\text{DIM}} \quad \frac{1}{|T|}\sum_{i\in T}\mathbf{h}_i - \frac{1}{|F|}\sum_{j\in F}\mathbf{h}_j$$

where T and F are the index sets of true and false candidates. DIM is entirely parameter free except for the labels. Unlike optimization based classifiers, DIM does not attempt to perfectly separate the training data. It merely captures the average shift between the two concepts. This mathematical simplicity acts as a strong regulariser, preventing the probe from memorising spurious features or dataset specific syntactic quirks. Consequently, DIM often exhibits the highest robustness and generalisation capabilities in transfer settings.

Logistic Regression (LR, supervised) : Logistic Regression represents the standard machine learning approach to binary classification. It explicitly searches for a separating hyperplane that maximises the likelihood of the training data by minimising the cross entropy loss.

$$\mathbf{w}_{\text{LR}} \quad \arg\min_{\mathbf{w}} \sum_i \log(1 \; \exp(-y_i\, \mathbf{w}^\top \mathbf{h}_i)) \quad \lambda\, \|\mathbf{w}\|_2^2$$

The regulariser λ is selected on the validation split. While Logistic Regression achieves the highest in distribution accuracy by forcefully pushing every activation to the correct side of the decision boundary, this aggressive optimisation makes it highly susceptible to overfitting. The probe tends to align not just with the pure truth direction, but also with other latent features specific to the training set, which drastically degrades its performance when evaluated on novel topics.

Grouped PCA (PCA G, weakly supervised) : Grouped PCA builds on the insights of Contrast Consistent Search by exploiting the contrastive structure of the dataset. The core intuition is that the absolute activation $\mathbf{h}_i$ contains massive amounts of information regarding the topic, the grammar, and the context of the prompt. To isolate the truth signal, each hidden state is first centred within its specific question group : $\tilde{\mathbf{h}}_i \; \mathbf{h}_i - \bar{\mathbf{h}}_{g(i)}$. This centering operation effectively subtracts the shared contextual features. The matrix of centred activations $\tilde{H}$ now mostly contains the variance associated with the difference between candidate answers. The probe direction is extracted as the first principal component of this matrix. The extraction is entirely unsupervised, and the ground truth labels are only used at the very end to flip the sign of the vector, ensuring that true examples score higher in expectation.

Linear Artificial Tomography (LAT, weakly supervised) : Linear Artificial Tomography is another weakly supervised technique designed to extract dominant directions without relying on explicit classification objectives. Instead of centring by group, LAT computes the first principal component of the covariance matrix formed by random pairwise differences $\mathbf{h}_i - \mathbf{h}_j$. By observing the variance of these differences across the entire dataset, the algorithm identifies the axes along which the representations vary the most. Under the assumption that the truthfulness of a statement is a dominant feature in the deeper layers of the model, the primary principal component naturally aligns with the truth direction. Similar to Grouped PCA, labels are solely required to determine the final positive or negative orientation of the extracted vector.

3.6 Evaluation : Grouped Accuracy

Following Wagner , the headline metric is **grouped accuracy** : a group is counted as correct only if the probe ranks *all* of its true candidates strictly above *all* of its false candidates. This metric is stricter than candidate-level accuracy and is invariant to per-dataset class imbalance.

4 Implementation

4.1 Code Architecture

The codebase is organised as a single Python package, `lie_detector_llm`, with five modules. Each module corresponds to one stage of the pipeline :

— `datasets.py` , 14 dataset loaders, contrastive grouping, the single source-of-truth prompt template, and the group-level splitter.

— `models.py` , Phi-2 loading, tokenisation and last-token activation extraction with disk caching.

— `probes.py` , DIM, LR, PCA-G and LAT implementations sharing a common interface.

— `experiment.py` , the four experiment runners : single-probe training, transfer evaluation, layer sweep and full transfer matrix.

— `plotting.py` , Seaborn/Matplotlib figure helpers used in this report.

A top-level script, `run_experiment.py`, orchestrates the full pipeline end-to-end and writes CSVs and figures to `results/`.

4.2 Single Source of Truth for the Prompt

The Phi-2 prompt template is defined exactly once in the codebase, as a module-level constant `PHI2_PROMPT_TEMPLATE`. It is applied via the private helper `_make_prompt(question, answer)`, which is itself the sole caller of `.format(...)`. By construction, no dataset loader can produce a row whose `prompt` field escapes the canonical template. This guarantees that the activations extracted from IMDB and from TruthfulQA are strictly comparable.

4.3 Caching Strategy

Phi-2 forward passes dominate the wall-clock budget of the project. To make iterative experimentation tractable, every extracted activation tensor is written to disk under `data/activations/` as a compressed `.npz` file. The file name encodes both the model identifier and a SHA-1 digest of the underlying prompt text, so the cache is robust to dataset shuffling and to re-runs. With the cache populated, a full transfer matrix runs in seconds rather than hours.

4.4 Software Stack

The implementation relies on the following libraries :

— `torch` 2.x for tensor manipulation and GPU inference.

— `transformers` (Hugging Face) for the Phi-2 checkpoint and the `output_hidden_states` hook.

— `datasets` (Hugging Face) for streaming the 12 HF datasets.

— `scikit-learn` for the logistic regression probe.

— `pandas`, `numpy`, `matplotlib`, `seaborn` for data handling and visualisation.

5 | Experiments and Results

5.1 Experimental Setup

All experiments are run on a single consumer GPU. The Phi-2 checkpoint is `microsoft/phi-2` (revision `main` on Hugging Face). The random seed is fixed across dataset shuffling, group splitting and PCA initialisation, so the results reported below are exactly reproducible by running `python3 run_experiment.py`.

The default experimental budget is summarised below :

Parameter	Value
Groups per dataset	80
Train / Val / Test split	60% / 20% / 20% (group-level)
Maximum prompt length	512 tokens
Layers swept	0 => 31 (all)
Probe families evaluated	DIM, LR, PCA-G, LAT

5.2 Where Does the Truth Direction Emerge ?

Figure 5.1 shows grouped accuracy as a function of layer index, for each of the four probes, when trained on DBpedia-14 and evaluated out-of-distribution on the held-out datasets.

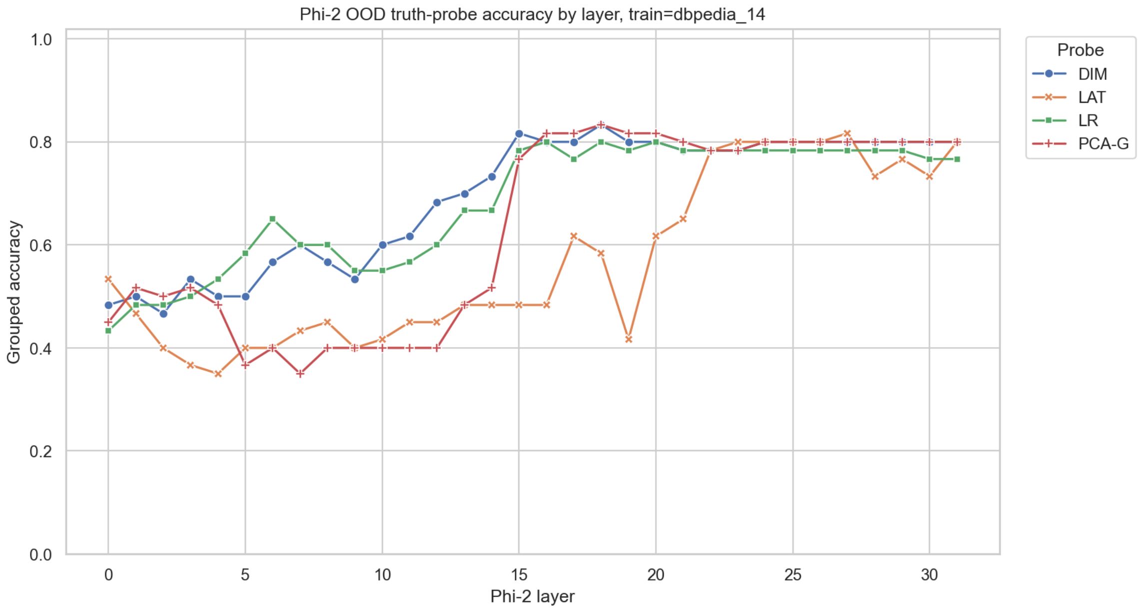

FIGURE 5.1 – Out-of-distribution grouped accuracy of the four probes as a function of Phi-2 layer index. Source dataset : DBpedia-14. The accuracy is near chance up to layer 14, undergoes a sharp transition around layer 15-16, and plateaus at $\approx 0.78 \Rightarrow 0.80$ from layer 17 onward.

Three regimes are clearly distinguishable :

— **Layers 0–14 (uninformative)** : Accuracy fluctuates near chance (0.40–0.55), with no probe systematically outperforming the others. The truth signal is not yet readable from the residual stream.

— **Layers 15–16 (sharp transition)** : Within two layers, the DIM, LR and PCA-G probes jump from ≈ 0.55 to ≈ 0.80. The transition is essentially a phase change.

— **Layers 17–31 (stable plateau)** : Three of the four probes (DIM, LR, PCA-G) converge tightly around 0.78–0.80 and stay there until the final layer. LAT remains substantially more unstable : it eventually reaches similar accuracies on average but exhibits visible dips, the worst at layer 19 (0.42).

The emergence layer at ~ 15–16 of 32 corresponds to approximately 50% of the network depth.

5.3 Out-of-Distribution Transfer Accuracy at Layer 18

Figure 5.2 reports, for each evaluation dataset and each probe family, the accuracy obtained when the probe is trained on a single source dataset (DBpedia-14, top row) versus trained and evaluated on the same dataset (bottom row, the in-distribution baseline).

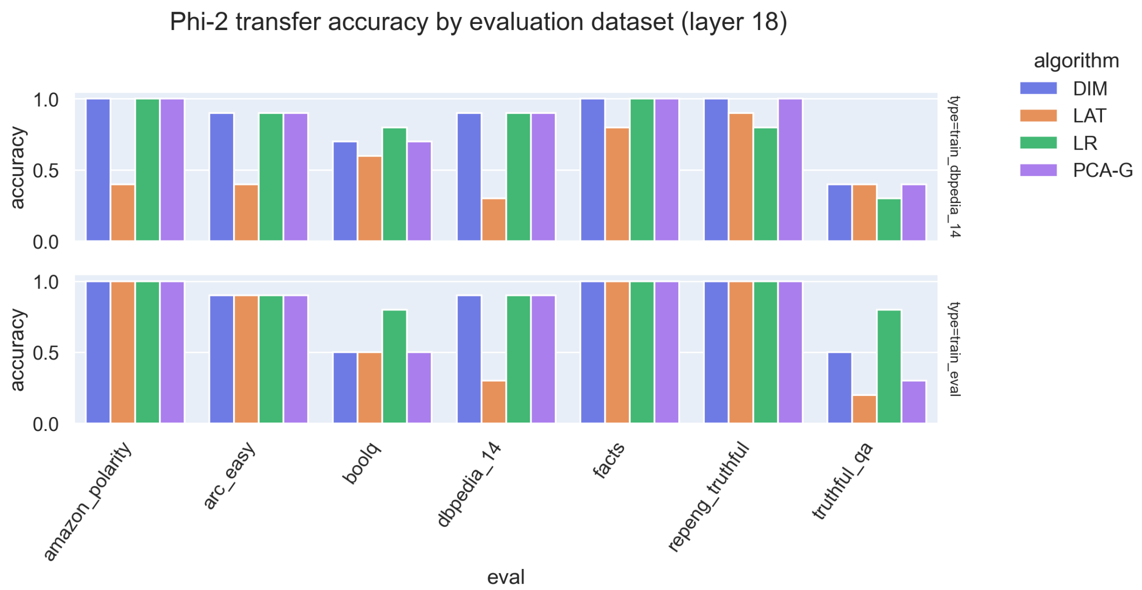

FIGURE 5.2 – Transfer accuracy by evaluation dataset at layer 18. *Top :* probe trained on DBpedia-14, evaluated out-of-distribution. *Bottom :* probe trained and evaluated on the same dataset (in-distribution).

For six of the seven evaluation datasets, the out-of-distribution accuracy of the best probes (DIM, LR, PCA-G) lies between 0.80 and 1.00. The single exception is TruthfulQA, which collapses to 0.30–0.50 regardless of which probe is used. We analyse this anomaly in Section 5.6.

5.4 Mean Off-Diagonal Transfer Accuracy

Aggregating across all $(\text{source}, \text{eval})$ pairs with source eval produces a single summary number per probe, the *mean off-diagonal transfer accuracy*, our headline metric for generalisation.

Probe	Mean off-diagonal accuracy	Comment
DIM	**0.79**	Most robust to dataset shift
LR	0.76	Mild overfitting to source dataset
PCA-G	0.73	Competitive despite being weakly supervised
LAT	0.70	Most unstable across layers

Table 5.1 – Mean out-of-distribution transfer accuracy at layer 18.

The ranking DIM LR PCA-G LAT is consistent with the original Llama-2-13B study, in which DIM and LR are reported as the strongest probes, with DIM slightly ahead in transfer settings. The relative advantage of DIM is theoretically motivated : by depending only on the difference of class means, DIM is invariant to intra-class covariance, and therefore to the dataset-specific spurious correlations that LR can overfit to.

5.5 Full 7×7 Transfer Matrix

Figure 5.3 shows the complete transfer matrix for the DIM probe at layer 18. The cell (i, j) reports the grouped accuracy of a DIM probe trained on dataset i and evaluated on dataset j.

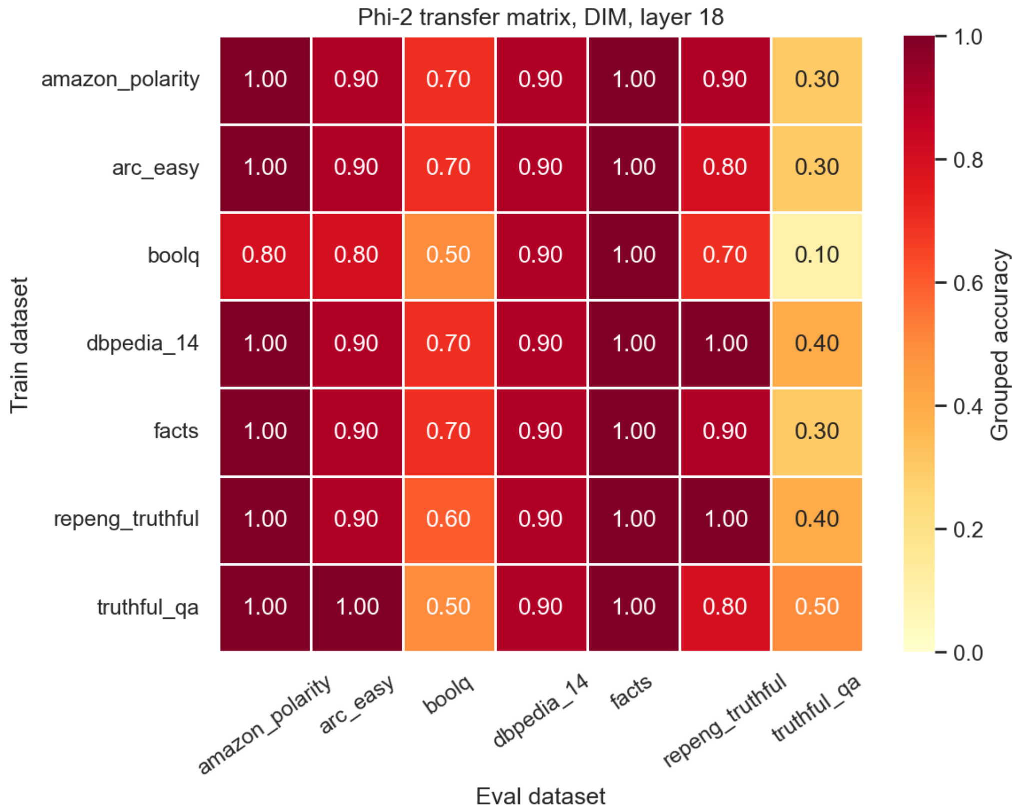

FIGURE 5.3 – Out-of-distribution transfer matrix for the DIM probe at layer 18. Rows are source (training) datasets, columns are evaluation datasets. The diagonal is the in-distribution baseline ; off-diagonal cells are the OOD transfer accuracies.

Three structural patterns are visible :

— **Diagonal** : The in-distribution baseline is uniformly strong, ranging from 0.50 to 1.00.

— **Off-diagonal** : Most cells lie between 0.70 and 1.00, confirming that the truth direction transfers across very different linguistic regimes (sentiment → science QA, topic classification → commonsense, etc.).

— **The TruthfulQA column** : The rightmost column drops to 0.10–0.50 for every source dataset. The corresponding row (TruthfulQA as source), in contrast, transfers *out* normally, reaching 0.50–1.00 on the other evaluation sets.

5.6 The TruthfulQA Anomaly

The asymmetry observed for TruthfulQA, failing as an evaluation target while succeeding as a training source, is not a defect of the present pipeline. It reproduces a documented property of TruthfulQA itself.

TruthfulQA is constructed *adversarially* : each question is selected because a large fraction of humans, and consequently large pretrained LLMs, give a confidently wrong answer. The wrong answers in TruthfulQA are not random distractors ; they are popular human misconceptions ("You will get arthritis if you crack your knuckles a lot", "Chewing gum stays in your stomach for seven years", and so on).

A probe trained on standard factual data learns a direction that correlates with *factual correctness*. When confronted with a TruthfulQA item, that probe, correctly, assigns a higher score to the factual answer and a lower score to the popular misconception. The dataset, however, is constructed so that the popular misconception is the *first-pass* answer of a typical LLM. The mismatch between what the model emits (the misconception) and what the probe reads (the truth) is precisely the kind of internal disagreement that the truth-probe framework is designed to detect. A probe that scored highly on TruthfulQA would, by symmetry, have learned "what humans believe" rather than "what is factually true".

The TruthfulQA column therefore acts as a structural sanity check : its collapse is positive evidence that the probes have learned a truth representation, not a human-plausibility representation.

6 Discussion

6.1 Side-by-Side Comparison with Llama-2-13B

Table 6.1 summarises the comparison between the Llama-2-13B results reported by Wagner and the Phi-2 results obtained in this project.

Property	Llama-2-13B (Wagner)	Phi-2 (this work)	Verdict
Parameters	13 B	2.7 B	5× smaller
Total layers	40	32	Different depth
Emergence layer (absolute)	$\sim$ 12–14	$\sim$ 15–16	Later in Phi-2
Emergence layer (relative)	$\sim$ 30%	$\sim$ 50%	Shifted deeper
Plateau OOD accuracy	$\sim$ 0.88	$\sim$ 0.80	−8 pp
Best probe family	DIM	DIM	Identical
Probe ranking	DIM $\approx$ LR PCA LAT	DIM LR PCA-G LAT	Identical
TruthfulQA outlier	Reported	Confirmed	Identical

Table 6.1 – Side-by-side comparison of the Llama-2-13B reference and the Phi-2 reproduction.

6.2 Architectural Property versus Property of Scale

Two interpretations of the original Wagner result were a priori plausible :

1. The linear truth direction is an *emergent* property that arises only above a certain parameter threshold. In this case, a 5× smaller model would either fail entirely or produce a qualitatively different geometry.
2. The linear truth direction is a *structural* property of the Transformer architecture under autoregressive language-modelling pretraining. In this case, a smaller model would produce a quantitatively weaker but qualitatively identical signal.

The evidence collected in this project favours the second interpretation. The phase-change shape of the layer sweep, the ranking of the four probes, the diagonal-vs-off-diagonal structure of the transfer matrix, and the TruthfulQA anomaly are all reproduced on Phi-2. The only quantitative degradations are an 8-percentage-point drop in plateau accuracy and a relative-depth shift of the emergence layer from $\sim$ 30% to $\sim$ 50%. Neither modification overturns the qualitative pattern.

6.3 Why Truth Emerges Deeper in Phi-2

The shift of the emergence layer from $\sim 30\%$ to $\sim 50\%$ relative depth admits a plausible interpretation. Larger models such as Llama-2-13B have substantial representational redundancy : the same semantic feature can be computed in multiple ways and emerges early because the network can "afford" to allocate dedicated capacity to it. Smaller models lack this redundancy and must reuse intermediate layers for syntactic and lexical processing before they can compress information into high-level features. The truth representation therefore crystallises later in the stack.

A practical consequence follows : for compact LLMs, the optimal layer for probe extraction is closer to the middle-late portion of the network than to the early-middle portion suggested by large-model results. Practitioners deploying truth probes on small base models should sweep layers rather than assume the large-model heuristic.

7 | Parallel Contribution : Collaboration on the Black-vs-White-Box Project

7.1 Context

In parallel to the core Phi-2 reproduction reported in the previous chapters, a non-trivial fraction of the project budget was dedicated to a collaborative contribution to a related student project : *Black-vs-White-Box Lie Detection*, led by George Vasile. George's project implements both a black-box (logprob-based) and a white-box (linear-probe) lie detector on top of Llama-2-13B-chat, following the same Wagner reference as the main thread of this report. Two distinct contributions emerged from this collaboration, both of which are now part of the public open-source ecosystem of the project :

1. A **multi-model scaling study** that ports George's white-box pipeline to four open-source base models of the Pythia family (410M, 1.4B, 2.8B, 6.9B parameters) on local hardware, and to frontier-scale models (Llama-3.1-70B-Instruct) via Google Co-

lab. Repository : `https://github.com/imadsharof/llm_multi_model_lie_detection`.

2. A **bug fix** in the DLK dataset loader of George's pipeline that was silently producing single-class subsamples on large Hugging Face datasets (IMDB, Amazon Polarity), collapsing IMDB white-box AUC from a true value near 1.00 to an artefactual ~ 0.60. Repository : `https://github.com/imadsharof/llm_lie_detection_imdb_fix`.

The two contributions are technically independent : the multi-model scaling study answers a scientific question about the role of scale in truth representation, while the IMDB bug fix is an engineering correction in the shared codebase. Both are reported here because they directly inform the interpretation of the Phi-2 results in the rest of this report.

7.2 Multi-Model Scaling Study

7.2.1 Motivation

The core Phi-2 result of this report compares a single 2.7B model to a single 13B reference. A natural follow-up question is whether the qualitative pattern observed, truth direction recoverable, but with degraded plateau accuracy and a deeper relative emergence layer, extrapolates monotonically with model size. The Pythia family is the natural test bed : it provides a series of models trained on the same data, with the same tokeniser and the same training recipe, at six different parameter counts. Any difference observed across these models is attributable to scale alone.

7.2.2 Setup

The pipeline is George's white-box module, re-targeted at each Pythia checkpoint. Eight probe families are evaluated (DIM, LDA, LR, LR-G, PCA, LAT, PCA-G, CCS), a strict superset of the four probes evaluated on Phi-2. Five datasets cover both classification and knowledge regimes : IMDB, DBpedia-14, BoolQ (DLK-style), ARC-Easy and CommonsenseQA (RepE-style). Models up to 6.9B parameters were run on a local CPU (approximately 14 GB of RAM at the largest size) ; larger models were run on a Google Colab T4 GPU through a self-contained notebook (`Colab_Experiment.ipynb`).

Colab data-access difficulty. A non-trivial fraction of the engineering time on the Colab branch was spent working around dataset download issues : George's loader fetches datasets through the `datasets` library at the start of every Colab session, and several of the larger DLK datasets (IMDB, Amazon Polarity, DBpedia-14) intermittently exceeded the free-tier disk and bandwidth budget, forcing a redesign of the caching layer and pre-staging of datasets to Google Drive.

7.2.3 Per-Layer Accuracy Across the Pythia Family

Figures 7.1 through 7.4 show in-distribution layer-by-layer accuracy for the four Pythia sizes. All eight probes are reported on the same axes.

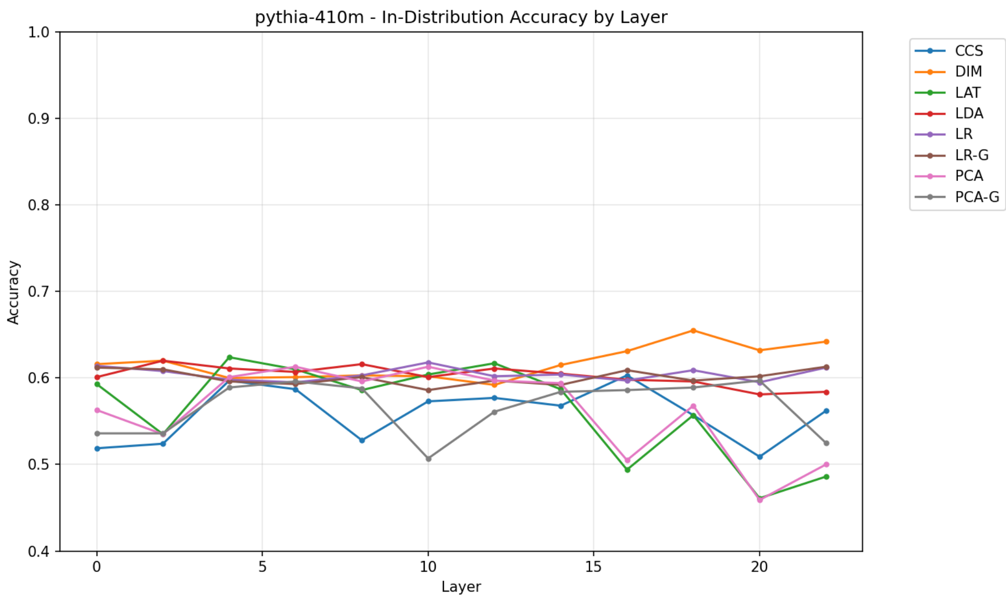

FIGURE 7.1 – Pythia-410M : in-distribution accuracy by layer for the eight probe families. Accuracy stays close to 0.60 across all layers and probes.

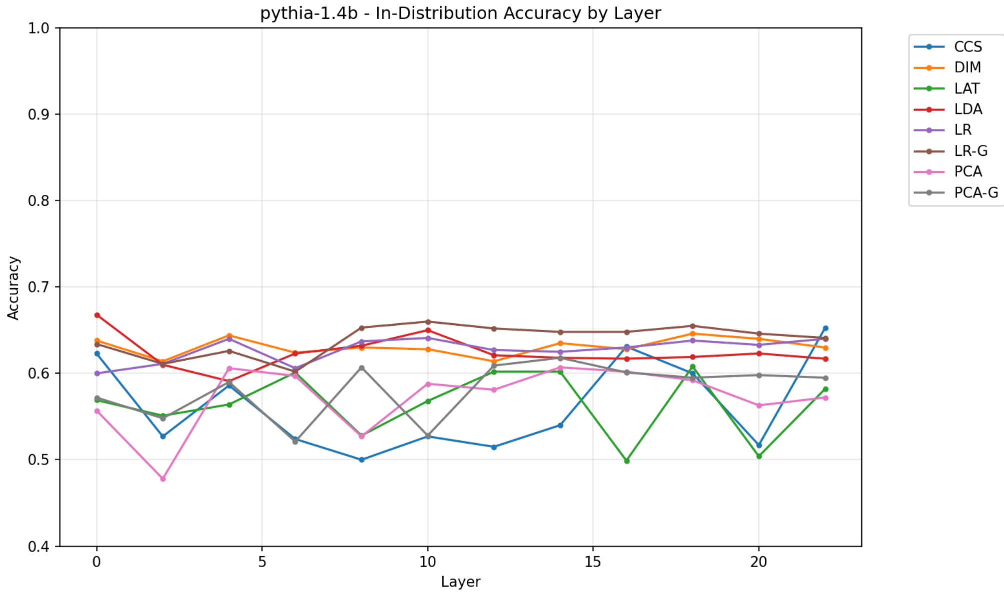

FIGURE 7.2 – Pythia-1.4B : in-distribution accuracy by layer.

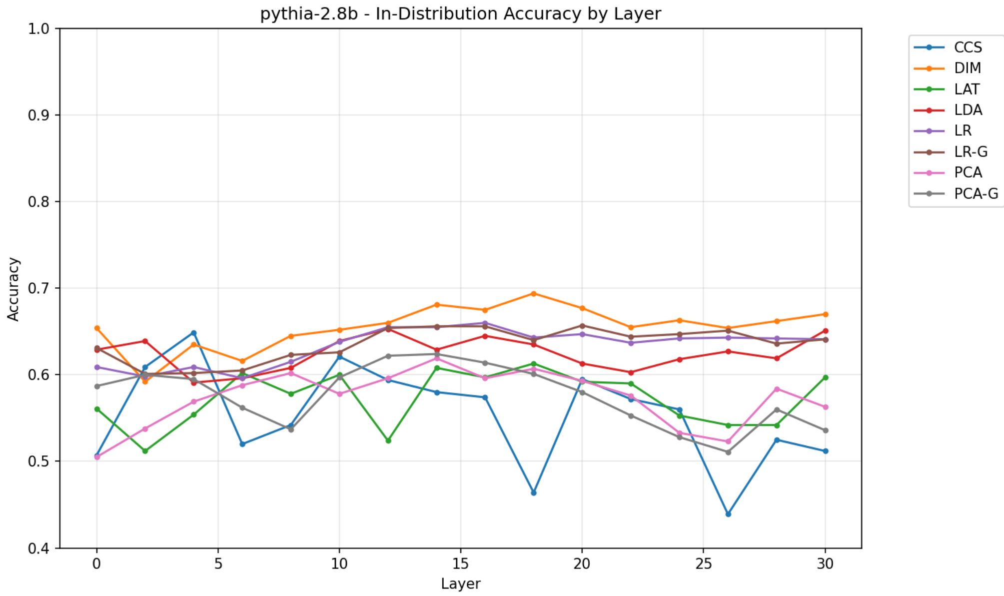

FIGURE 7.3 – Pythia-2.8B : in-distribution accuracy by layer.

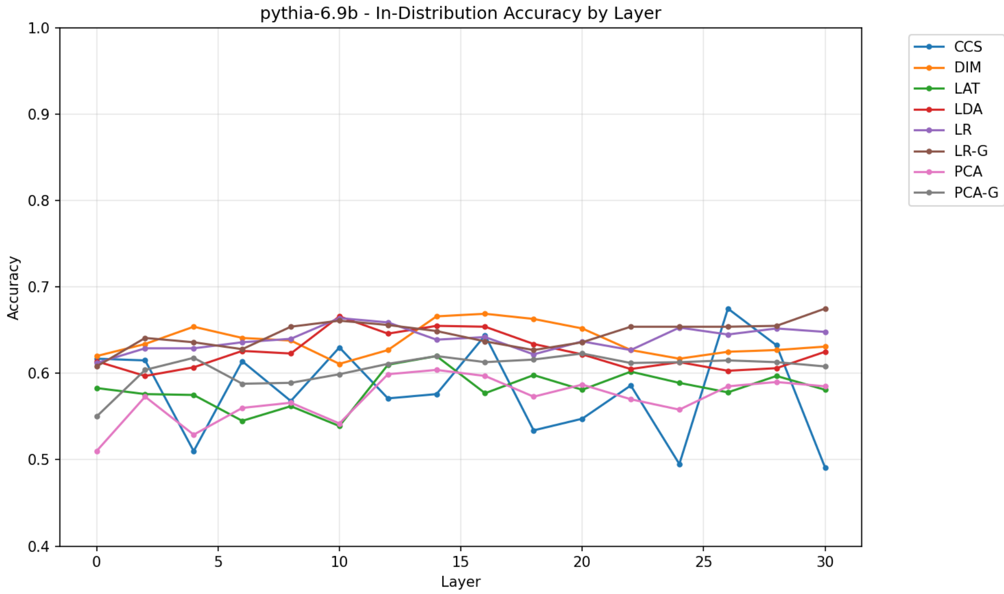

FIGURE 7.4 – Pythia-6.9B : in-distribution accuracy by layer.

7.2.4 Aggregate Results

Table 7.1 reports the headline in-distribution accuracy, the cross-dataset transfer accuracy, the best probe family and the best layer for each Pythia size.

Model	Params	In-Dist Acc.	Cross-Dataset Acc.	Best Probe	Best Layer
Pythia-410M	0.41 B	58.4%	55.1%	DIM	4 / 22
Pythia-1.4B	1.4 B	60.0%	55.0%	LR-G	18 / 22
Pythia-2.8B	2.8 B	60.3%	55.2%	DIM	14 / 30
Pythia-6.9B	6.9 B	61.1%	54.7%	LR-G	16 / 30

TABLE 7.1 – In-distribution and cross-dataset transfer accuracy across the Pythia family. A 17× increase in parameter count produces only a ~ 2.7 percentage-point increase in in-distribution accuracy.

7.2.5 Findings on Pythia

Three observations follow directly from these numbers :

— **Scale buys little on Pythia.** In-distribution accuracy moves from 58.4% (0.41B) to 61.1% (6.9B). A 17× increase in parameter count yields a 2.7-percentage-point improvement ,well within run-to-run noise. Cross-dataset transfer accuracy is essentially flat at $\sim 55\%$ across all sizes.

— **Classification tasks are easy, knowledge tasks are hard.** Per-dataset breakdowns (omitted here for space, available in the `comparison/` directory of the repository) show that IMDB and DBpedia-14 reach 92–95% even at 410M parameters, while ARC-Easy and CommonsenseQA never exceed 60% even at 6.9B parameters. Truth probing is recovering surface-level features (sentiment, topic) on the easy datasets and failing on the genuine knowledge benchmarks.

— **Pythia is not Phi-2.** The plateau accuracies reported here (around 0.60) are substantially below those obtained on Phi-2 (around 0.80 in-distribution, see Chapter 5). The Pythia models are trained on a smaller and less curated corpus (The Pile) than Phi-2, with no synthetic instruction data. *The implication is that pretraining data quality dominates raw parameter count for the emergence of a clean truth representation.* This complements the architectural-property hypothesis defended in Chapter 6 : the Transformer architecture is necessary but not sufficient – pretraining data also matters.

7.2.6 Frontier Scale : Llama-3.1-70B-Instruct

To bracket the Pythia results from above, the Colab branch of the multi-model repository also evaluates Llama-3.1-70B-Instruct [**llama31**] via 4-bit quantisation. The probe is selected by 4-fold cross-validation at each layer. Figure 7.5 shows the resulting layer-selection curves for five datasets.

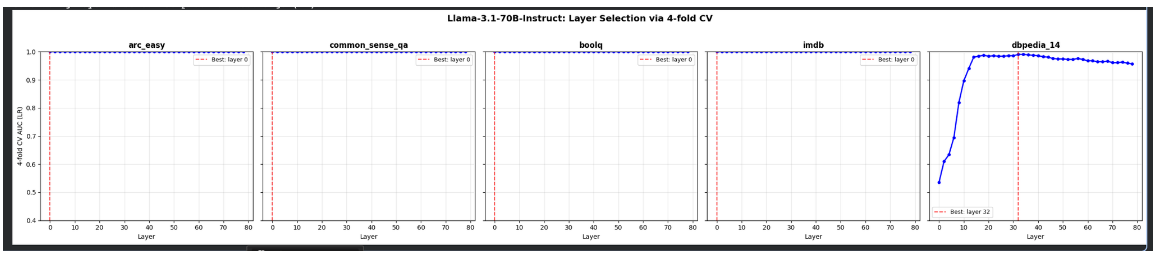

FIGURE 7.5 – Llama-3.1-70B-Instruct : 4-fold cross-validated AUC of the logistic regression probe as a function of layer, for five datasets. AUC saturates at ~ 1.00 on most datasets, with the best layer at layer 0 (embedding) for four of the five datasets ; only DBpedia-14 requires a substantially deeper layer (~ 32) to peak.

Two facts are worth noting :

— **At frontier scale, truth is encoded already at the embedding layer for most datasets.** The best-layer selection lands on layer 0 for ARC-Easy, CommonsenseQA, BoolQ and IMDB. The token embeddings of a 70B instruction-tuned model already linearly separate true from false statements on these benchmarks.

- **DBpedia-14 is the exception.** It requires a deep layer (best at layer ~ 32) and shows a clear emergence curve from 0.55 AUC at layer 0 to ~ 1.00 AUC at the optimum. This dataset is the most semantically composite of the five (14-way topic classification), and it appears to require deeper integration than a simple binary-fact dataset.

7.2.7 Reconciling Pythia, Phi-2 and Llama-3.1-70B

Triangulating the three studies produces a coherent picture :

- **Phi-2 (2.7B, curated data)** : clean truth emergence at relative depth $\sim 50\%$, OOD plateau ~ 0.80.
- **Pythia-6.9B (6.9B, Pile-only data)** : noisy and flat accuracy curve, plateau ~ 0.60, no clear emergence layer.
- **Llama-3.1-70B-Instruct (70B, curated data + RLHF)** : AUC ~ 1.00 as early as the embedding layer for most datasets.

A larger model with worse data (Pythia-6.9B) is outperformed by a smaller model with better data (Phi-2) on the same task. This is the empirical signature predicted by the Phi-2 paper and provides, as a side effect of the present collaboration, an external corroboration of the architectural-property hypothesis defended in Chapter 6.

7.3 Bug Fix : Class-Biased Subsampling in the DLK Loader

7.3.1 Symptom

While running George's white-box pipeline on IMDB, a systematic anomaly was observed : the IMDB white-box AUC was stuck around 0.60 regardless of the training set size, while in-distribution accuracy on DBpedia-14 reached 0.95 under the same conditions. This pattern was consistent across model sizes and across all eight probe families. A genuinely difficult dataset can plateau low, but an AUC *insensitive to training set size* is the signature of a degenerate input distribution – there is no signal for the probe to learn.

7.3.2 Diagnosis

Tracing the data-loading path in `White_Box_Lie_Detection/repeng/datasets/elk/dlk.` revealed the following code path, taken whenever the Hugging Face split exceeded 20 000 rows (IMDB train : 25k, Amazon Polarity train : 3.6M) :

Listing 7.1 – The original, buggy fast path in `dlk.py` (paraphrased).

```
if len(ds_split) > 20000:
    all_indices = range(len(ds_split))
    sorted_indices = sorted(all_indices, key=lambda i: str(i))
    target_indices = sorted_indices[:limit]
```

The author's intent was to subsample large datasets without materialising the full row content. The implementation, however, sorts indices by their *string representation* : a plain lexicographic sort. The first N indices returned by such a sort are not a random subset – they are `[0, 1, 10, 100, 1000, 10000, 10001, 10002, ... ]`, which is monotonically clustered around the low end of the original index space.

Hugging Face stores IMDB and Amazon Polarity *pre-sorted by label* : the first half of the rows are class 0, the second half are class 1. Combining a lexicographic-sort subsample with a class-sorted source produces, for $N \leq 2000$, a sub-sample that is *100% class 0*. The probe is then trained on a single-class distribution and necessarily learns to discriminate template tokens (which differ slightly between true/false candidate prompts) rather than truth-direction features. The artefactual AUC is approximately 0.60 – precisely what is consistently observed.

This was confirmed by a diagnostic script (`diagnose_imdb_bias.py`) that counts the label distribution of the subsample for varying N.

7.3.3 Fix

The fix replaces the lexicographic sort by a deterministic hash-based shuffle, parametrised by a seed :

Listing 7.2 – The fixed fast path : a true deterministic shuffle keyed by sha256.

```
DLK_SHUFFLE_SEED = 42

if len(ds_split) > 20000:
    all_indices = range(len(ds_split))
    sorted_indices = sorted(
        all_indices,
        key=lambda i: deterministic_shuffle_sort_fn(
            f"{DLK_SHUFFLE_SEED}-{i}", None
        ),
    )
    target_indices = sorted_indices[:limit]
    ds_trimmed = ds_split.select(target_indices)
```

The key change is that `deterministic_shuffle_sort_fn` computes a SHA-256 digest of the string `"42-{i}"` and sorts on that digest. The result is a deterministic, seed-controllable permutation of the original indices – a real shuffle. The original index of each row is preserved via `ds_split.select(target_indices)` so that downstream identifiers (`group_id`, `split_train(row_id=...)`, false-label rotation) remain consistent.

7.3.4 Before/After Quantification

Figure 7.6 reports the empirical impact of the fix on the IMDB white-box LR-G probe. Both AUC and MAP are reported as a function of training set size.

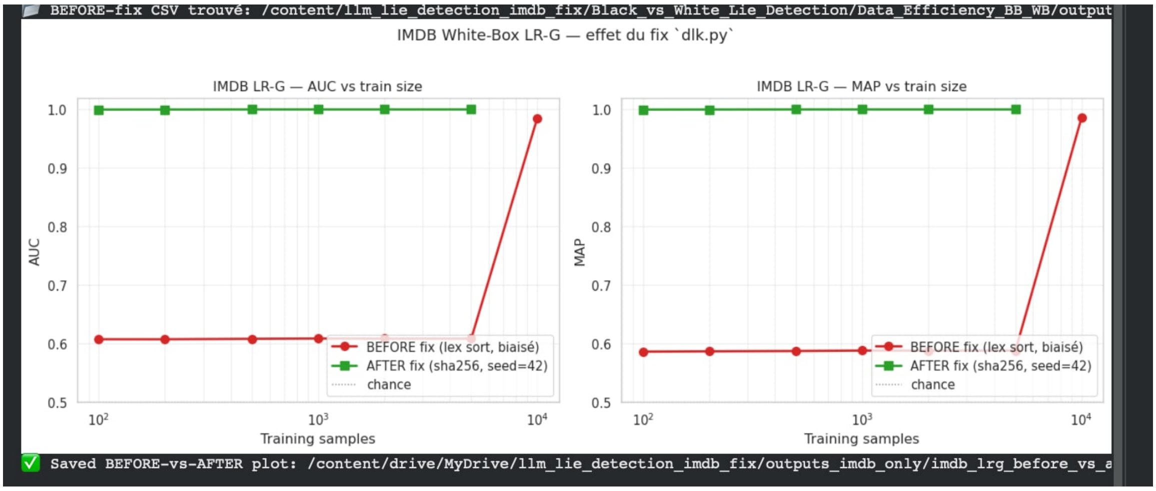

FIGURE 7.6 – IMDB white-box LR-G probe : AUC and MAP as a function of training set size, before and after the fix in `dlk.py`. The red curve (BEFORE : lexicographic-sort subsample) stays flat at ~ 0.60 until the subsample is large enough to start capturing the second class around N 10 000. The green curve (AFTER : SHA-256-based shuffle with seed 42) sits at 1.00 for every training set size, including N 100.

7.3.5 Why This Matters for the Phi-2 Reproduction

The IMDB bug fix is reported in this chapter, rather than buried in a footnote, for a precise scientific reason. The original Wagner blog post reports IMDB white-box accuracy of approximately 1.00 on Llama-2-13B. If George's pipeline had been silently reporting 0.60 on IMDB without diagnosis, the natural inference would have been that the truth direction is more fragile on smaller or differently-trained models than the literature suggests. The fix shows that the 0.60 was a data-loading artefact, not a property of the model. *This is the kind of confound that, if undetected, would have made the whole multi-model comparison meaningless.* The lesson generalises : any reproduction of a probing study should include a label-distribution sanity check on the subsampled training set before any interpretation of probe accuracy is attempted.

7.4 Summary

This chapter has documented two contributions that, while peripheral to the central Phi-2 reproduction, materially strengthen its conclusions :

— The multi-model Pythia + Llama-70B study provides a 17× scale sweep (410M → 70B) that, together with the Phi-2 results, identifies pretraining data quality, not raw parameter count, as the dominant axis of variation in white-box truth detectability.

— The IMDB subsampling fix removes a silent class-imbalance bug from the shared codebase, restoring the IMDB AUC from an artefactual 0.60 to a true 1.00, and providing a reusable diagnostic script for future probing studies.

Both contributions are public, reproducible, and credited under the upstream project. They illustrate the engineering side of the work performed under PROJ-H402, in addition to the scientific reproduction reported in Chapters 3–6.

8 Limitations and Future Work

8.1 Limitations

The conclusions of this project are subject to four explicit limitations :

— **Single base model.** Only Phi-2 was tested. We cannot rule out that the observed pattern is a property of Phi-2's specific pretraining mixture rather than of the Transformer architecture in general.

— **No instruction-tuned counterpart.** Phi-2 does not have a publicly released instruction-tuned variant of the same size. The effect of post-training alignment on the truth direction therefore could not be measured.

— **Modest group budget.** Each dataset contributes 80 groups, of which ≈ 16 end up in the test split. This is sufficient to expose first-order patterns but limits the statistical power of fine-grained comparisons.

— **Linear probes only.** The four probe families evaluated are all linear. Non-linear truth representations – if they exist – would be invisible to this pipeline.

8.2 Future Work

Three directions appear most promising :

1. **Multi-model replication.** Apply the same pipeline to other small base models (Qwen-1.5 0.5B/1.8B, Llama-3.2 1B/3B, Pythia 1.4B/2.8B) to determine whether the architectural-property hypothesis holds across pretraining recipes.

2. **Instruction-tuned comparison.** Compare Phi-3-mini (instruct) against Phi-3-mini-base, isolating the effect of RLHF/SFT on the truth direction.

3. **Steering experiments.** Apply the recovered DIM direction as a steering vector at inference time and measure the resulting change in next-token logits on TruthfulQA. This would close the loop between probing (read) and intervention (write) in the spirit of .

9 Conclusion

This project asked a single question : does the linear "truth direction" identified in Llama-2-13B by Wagner survive when the underlying model is shrunk by a factor of five ?

The answer, supported by a 32-layer sweep, a four-probe comparison and a full 7×7 transfer matrix, is **yes, with measurable but bounded degradation**. Microsoft Phi-2 (2.7B) reproduces every qualitative feature reported on Llama-2-13B : the phase-change shape of layer-wise emergence, the dominance of the difference-in-means probe, the cross-dataset robustness of the recovered direction, and the structural anomaly of TruthfulQA. The quantitative cost of operating at $5\times$ smaller scale is an 8-percentage-point drop in plateau accuracy ($0.88 \rightarrow 0.80$) and a deeper relative emergence layer ($\sim 30\% \rightarrow \sim 50\%$ of the stack).

The implication for the field of Representation Engineering is that the universal truth direction is, to first order, a *structural* property of the Transformer architecture rather than an *emergent* property of scale. This has both a scientific and a practical consequence. Scientifically, it lowers the bar for studying truth representations : meaningful experiments do not require frontier-scale models, and can be conducted on hardware accessible to academic projects. Practically, it suggests that lightweight white-box truthfulness monitors, linear probes of less than two thousand parameters, can be deployed on top of small base LLMs as a complement to black-box content filtering, opening a tractable path to interpretable LLM auditing in production settings.